

TextMap vs GeoJSON: how LLMs read maps

Preliminary results · real OpenStreetMap scenes · answers graded automatically against geometric ground truth · temperature 0 · each format shown in isolation

Setup. The same maps are given to the model in two representations — **GeoJSON** (raw geometry, the baseline an LLM agent normally receives) and **TextMap** (an ASCII grid + a measurement legend). The model answers 10 spatial questions per map; a geometry oracle grades every answer. Tokens, latency and hallucinated entity ids are logged per answer.

1. Main result — 60 maps, Claude Haiku 4.5

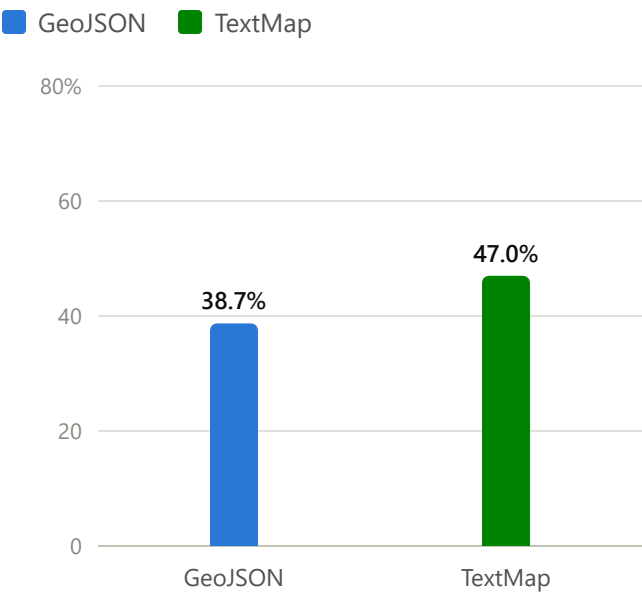


Figure 1. Accuracy over 1,200 graded answers (60 maps × 2 formats × 10 questions). Paired McNemar test: $\chi^2 = 14.46$, $p = 0.0001$. TextMap also used 33% fewer input tokens (6,739 vs 10,025 per call).

2. Scaling the model does not help GeoJSON

The same 20 maps, three model tiers. On raw GeoJSON, accuracy is flat — a stronger model extracts nothing more from raw coordinates. On TextMap, accuracy climbs with the model.



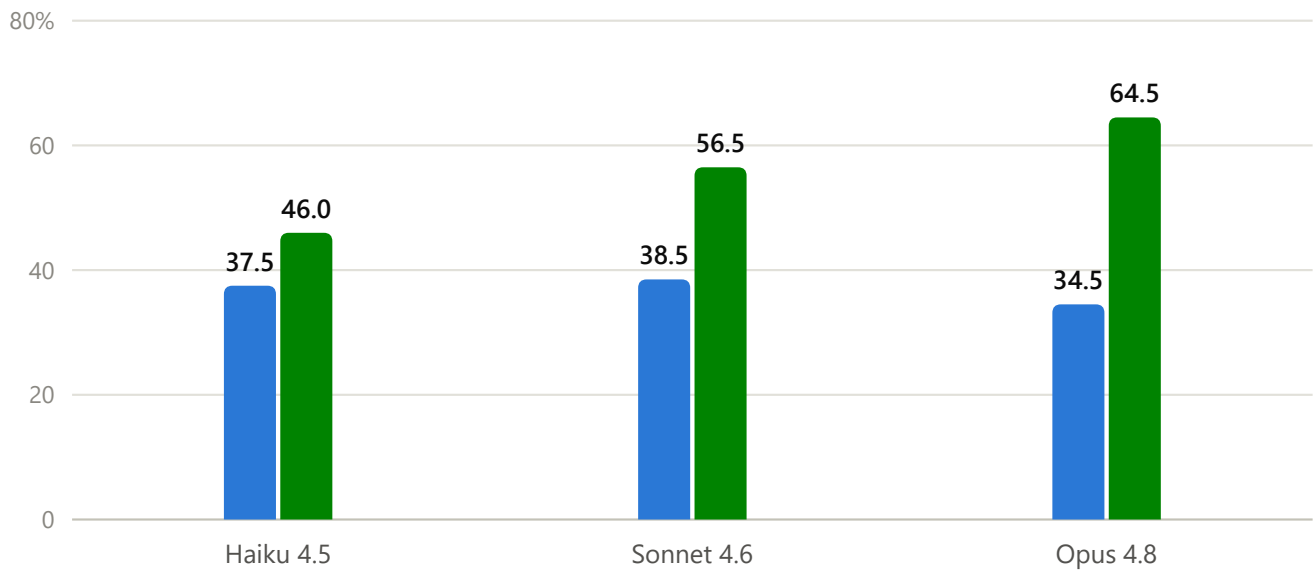


Figure 2. Accuracy (%) on the identical 20 maps for all three models. GeoJSON differences across models are within noise; the TextMap gain grows from +8.5 (Haiku) to +30 points (Opus). The representation, not the model, is the bottleneck.

3. Where TextMap wins — and where it does not

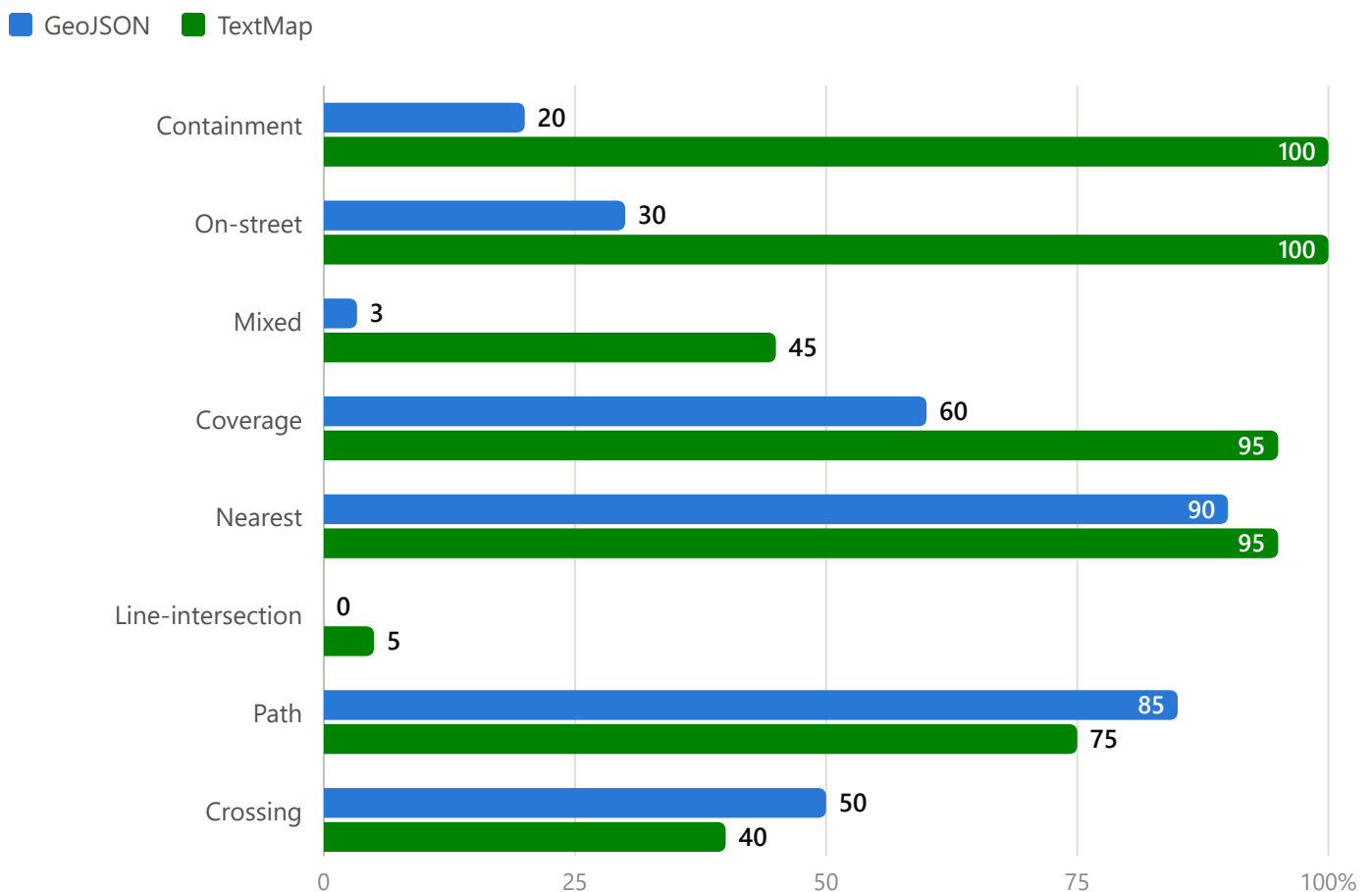


Figure 3. Accuracy (%) by question category, Opus 4.8, 20 maps, sorted by TextMap advantage. TextMap dominates spatial-relationship questions (containment, on-street, mixed, coverage); raw coordinates stay competitive on point-

distance questions (nearest, path) and line-crossing, a known limitation of the grid's cell resolution. Both formats fail line-intersection. Reported as-is.

4. Cost — TextMap stays compact as maps grow

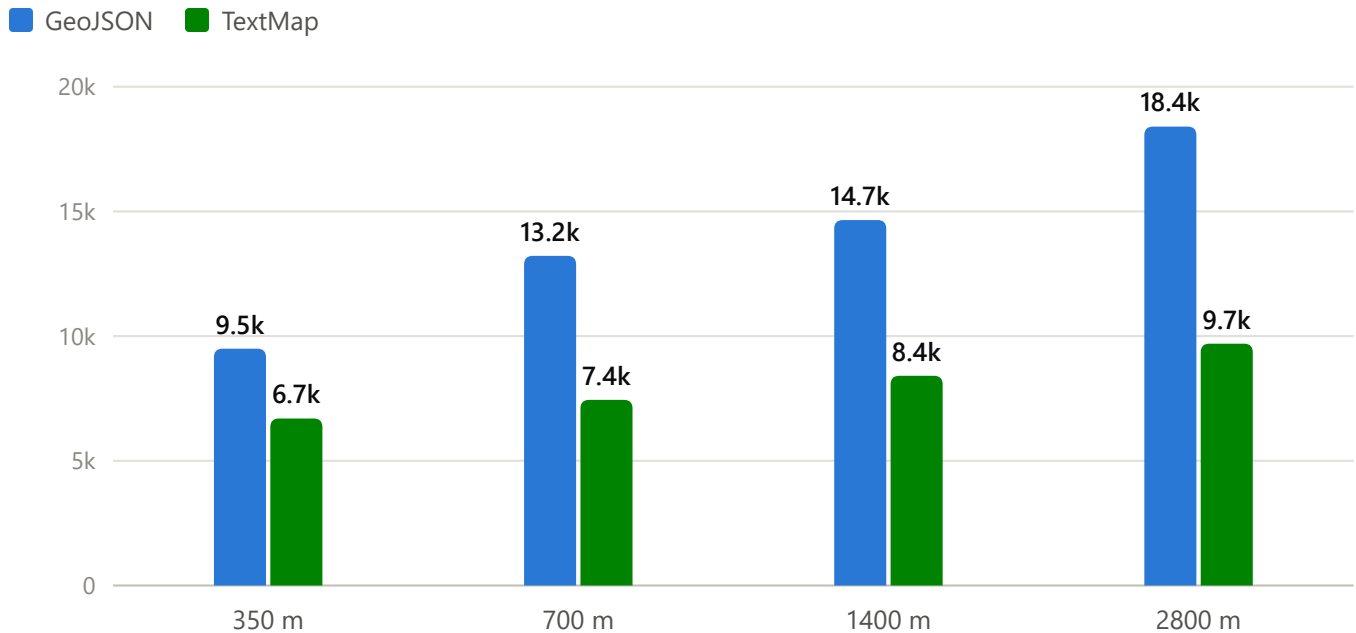


Figure 4. Average input tokens per call by map size (Haiku, same scenes at four area sizes). GeoJSON grows with the geometry; TextMap stays near-flat — at 2.8 km it uses 47% fewer tokens at equal accuracy.

5. Statistical significance (McNemar, paired)

Each map is answered in both formats, so the test counts only the answers where the formats disagree: b = TextMap right & GeoJSON wrong, c = the reverse.

Model	Maps	b	c	χ^2	p
Haiku 4.5	60	108	58	14.46	0.0001
Sonnet 4.6	20	49	13	19.76	<0.0001
Opus 4.8	20	64	4	51.19	<0.0001

On Opus, of the 68 answers where the formats disagreed, TextMap was right in 64.

6. Not yet proven — next runs

- **One neighbourhood:** all maps are distinct blocks within a ~2 km area of Manhattan (one street morphology). A second, differently-shaped city is next.
- **Held-out questions:** the question set was designed alongside the format; a separate question set the design never saw will rule out benchmark tuning.

- **Anthropic models only** so far; GPT / Gemini / open models are queued.
- Sonnet/Opus points are 20 maps each (the 60-map run is the certified headline).

Reliability note: both text formats almost never invent entities (hallucinated-id rate $\leq 1\%$). Rendering the same maps to an image raised it to $\sim 40\%$ — rasterization destroys entity identity. Full four-format comparison (incl. WKT, image) to follow.

Data: the attached CSVs contain one row per graded answer (columns: arm = format, correct, category, inputTokens). Every number above is reproducible from them.